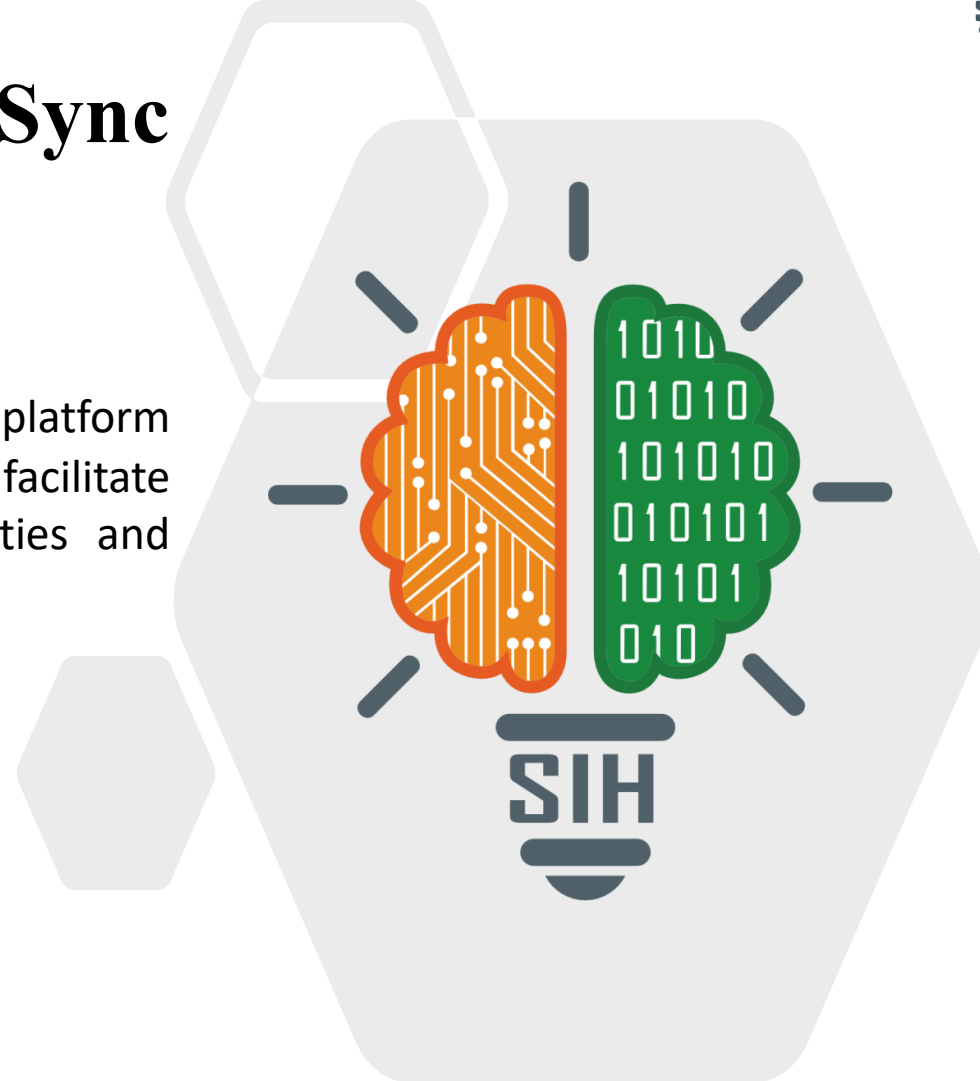




## CivicSync

- **Problem Statement ID – 26043**
- **Problem Statement Title-** A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Mind Mavericks**



## Problem

## Proposed Solution

❖ **Problems remain unstructured & fragmented**

Citizen complaints come from multiple channels, but similar issues are rarely clustered, prioritized, and converted into clear, actionable problems.

❖ **Expertise remains disconnected**

Universities, faculty, students, researchers, and industry have relevant expertise, but the right problem-solving team is difficult to identify and connect.

❖ **The solution journey is disconnected**

Problem discovery, solution development, implementation, and impact measurement often happen separately — creating a gap from **Problem** → **Expertise** → **Solution** → **Implementation** → **Impact**.

➤ **AI-Powered Challenge Intelligence**

CivicSync converts citizen reports into **structured, prioritized challenges**, identifies required expertise, and **clusters related issues**.  
**Understand** → **Cluster** → **Prioritize**

➤ **End-to-End Collaborative Problem Solving**

CivicSync connects challenges with **students, researchers, universities & industry** to develop, implement, and **track real-world impact**.  
**Problem** → **Expertise** → **Solution** → **Impact**

## Innovation &amp; Uniqueness



## 1. AI Problem Intelligence

Converts citizen reports into structured challenges



## 2. Problem Clustering &amp; Priority

Detects duplicate issues & prioritizes by impact



## 3. Explainable Capability Matching

Matches problems with students, universities & industry



## 4. Solution-to-Implementation

Connects validated solutions with startups & MSMEs



## 5. Evidence-Backed Impact

Tracks Baseline to Target to Actual with verification



## 6. Impact Passport &amp; Reputation

Transparent Impact Passport & verified reputation



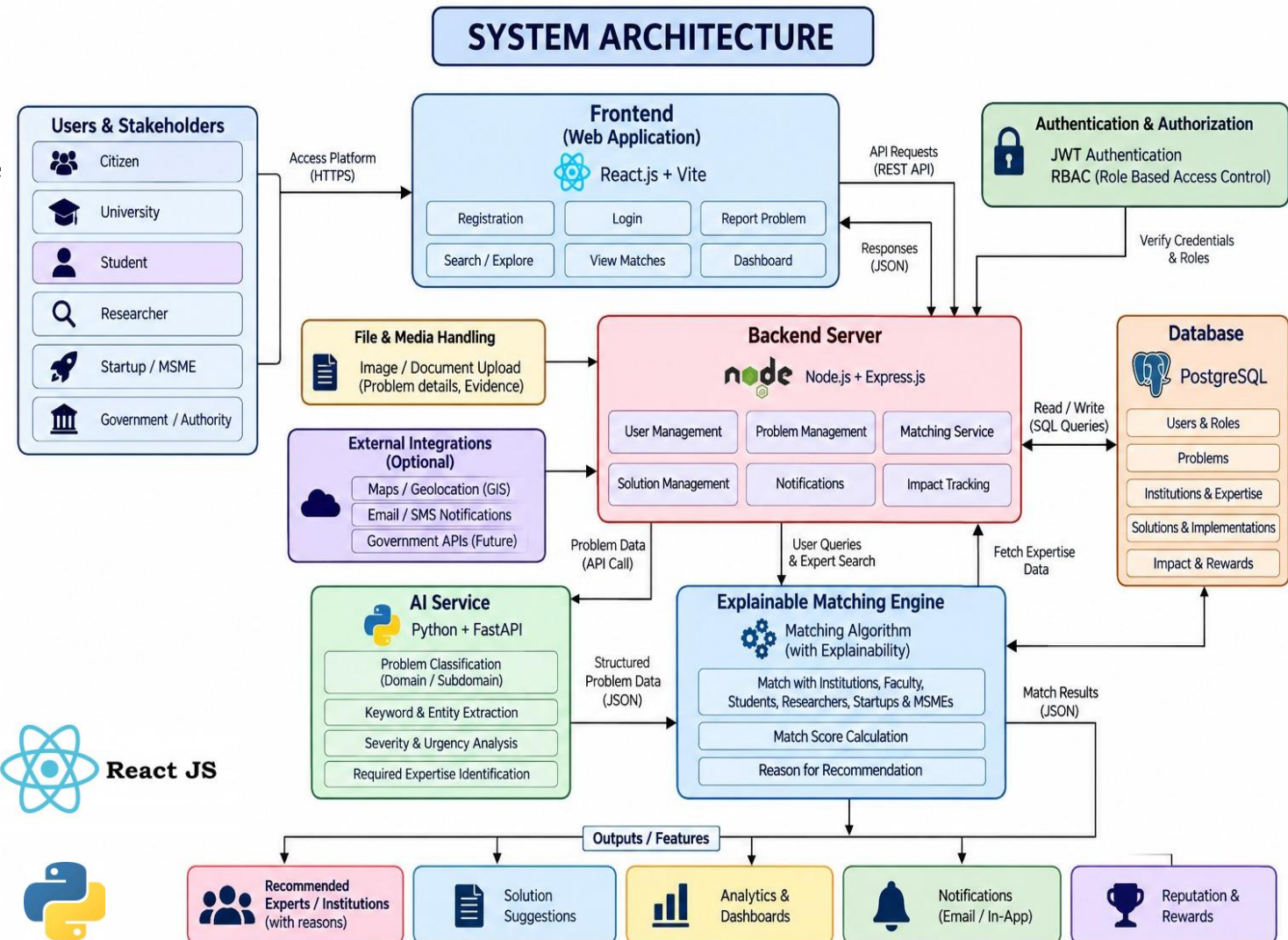
**CivicSync** does not stop at finding a problem or an expert — it creates a traceable path from  
**Problem** → **Solution** → **Implementation** → **Verified Impact**

## TECHNOLOGY STACK

- ❖ Frontend: HTML, Tailwind CSS, JavaScript, React.js
- ❖ Backend: Node.js, Express.js, REST APIs, bcrypt.
- ❖ RBAC(Role Based Access Control)
- ❖ Database: PostgreSQL
- ❖ Authentication: JWT Authentication
- ❖ AI/Intelligence: Python, FastAPI, Rule based NLP, Pattern Analysis, LLMs, Embeddings



## SYSTEM ARCHITECTURE





## FEATURES

1. **Citizen Problem Reporting :**

Citizens can submit real-world societal problems with location, category and affected population.

2. **AI Challenge Intelligence :**

AI identifies domain, subdomain, severity, urgency, keywords and required expertise from the problem.

3. **Explainable Expertise Matching :**

Matches problems with relevant institutions, faculty, students, researchers, startups and MSMEs with match scores and reasons.

4. **Duplicate & Similar Problem Detection :**

Identifies potentially duplicate or related problems to reduce repeated efforts.

5. **Implementation & Pilot Tracking :**

Tracks milestones, progress, evidence and blockers during solution implementation.

6. **Impact Measurement :**

Measures baseline vs actual outcomes, citizen feedback and verified impact.

7. **Reputation, Rankings & Rewards :**

Rewards verified meaningful contributions through reputation points, badges and rankings.

Technical Feasibility 

- Modular stack: React + Node.js + FastAPI + PostgreSQL — UI, workflow, DB, and AI cleanly separated
- MVP-ready AI: rule-based NLP + pattern matching — no GPU or massive datasets needed

Operational Feasibility 

- Human-in-the-loop: AI classifies & ranks; authorities validate before action
- Multi-role RBAC: tailored dashboards for Citizens, Students, Universities, MSMEs, Govt

Data Feasibility 

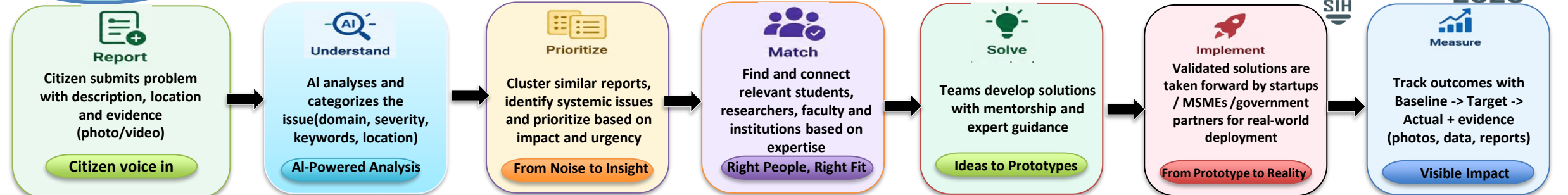
- Cold-start ready: bootstrapped with curated capability profiles for institutions, experts, startups
- Built for growth: structured schema, ready for future vector/LLM integration

Scalability & Sustainability 

- Phased rollout: District → State → National deployment
- Plug-and-play growth: new civic domains, AI models, GIS layers, and external registries

**VIABILITY FORMULA** Buildable Now → Low-Cost MVP → Human-Controlled → Modular Ecosystem → Scalable Architecture

**HONEST LIMITATION** MVP demonstrates rule-based AI with curated capability profiles — semantic vector search, fine-tuned LLMs, and real-time civic-portal API integration are scoped as Phase 2 extensions.



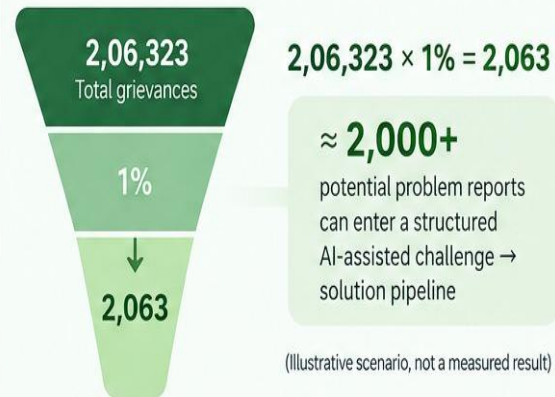
## 1. PROBLEM SCALE

CPGRAMS grievances received  
(Central Ministries + States/UTs, June 2025)

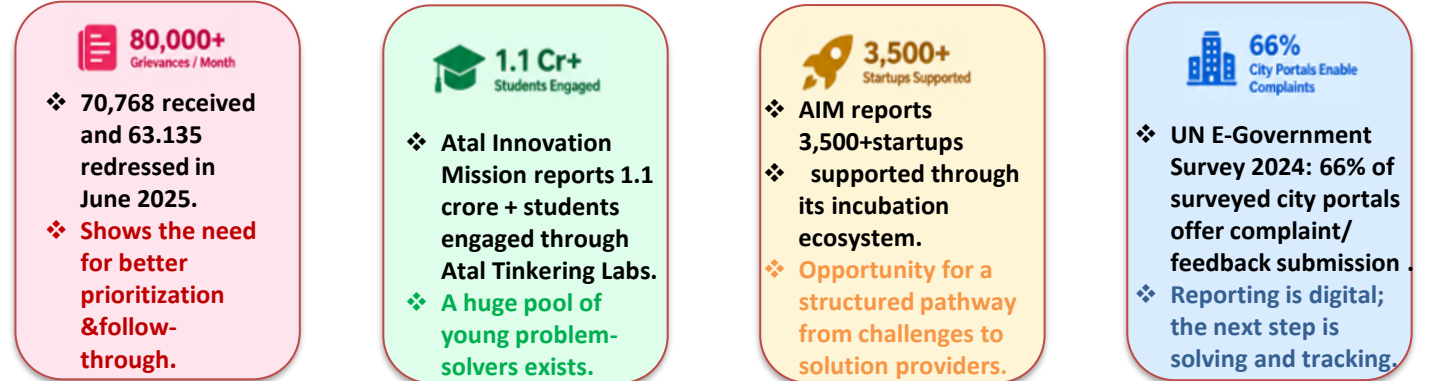
**2,06,323**

Real citizen problems  
across the country

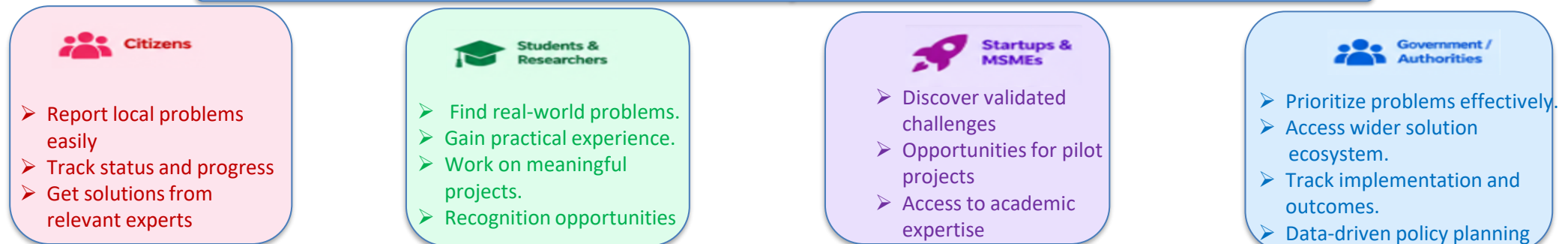
## 2. EVEN 1% MAKES A DIFFERENCE



## IMPACT



## Benefits for Key Stakeholders



**WHERE THE GAP EXISTS**

Existing platforms solve important parts, but not the complete journey.

| Capability                            | Existing Platforms<br>(CPGRAMS / MyGov / AIM-ARISE) | Our Platform<br>(CivicSync) |
|---------------------------------------|-----------------------------------------------------|-----------------------------|
| Citizen problem / grievance reporting | ✓                                                   | ✓                           |
| Structured problem intelligence       | ~                                                   | ✓                           |
| Automated problem clustering          | ~                                                   | ✓                           |
| Explainable expertise matching        | —                                                   | ✓                           |
| Implementation lifecycle tracking     | ~                                                   | ✓                           |
| Baseline → Target → Actual            | —                                                   | ✓                           |



= Core capability



= Limited / Not core capability



= Not a core capability

Expert Finding  
Systems: A  
Systematic Review

- ❖ **Evidence:** Review of 96 expert-finding studies; ~65% focused on academic environments. Expertise is commonly derived from profiles, documents and textual evidence.
- ❖ **CivicSync:** Capability-based matching of problems with relevant students, researchers, faculty and institutions.
- ❖ **Source:** <https://www.mdpi.com/2076-3417/9/20/4250>

Automated Citizen  
Request  
Classification

- ❖ **Evidence:** Research shows that large volumes of unstructured citizen requests can benefit from automated classification.
- ❖ **CivicSync:** AI-based problem intelligence for domain, severity, urgency and required expertise extraction.
- ❖ **Source:** <https://www.sciencedirect.com/science/article/pii/S0306457320309055>

Citizen Complaint  
Pattern Analysis

- ❖ **Evidence:** Research on citizen complaints identifies spatial and temporal patterns that can reveal recurring or concentrated issues.
- ❖ **CivicSync:** Related-problem clustering and creation of systemic/master challenges.
- ❖ **Source:** <https://www.sciencedirect.com/science/article/pii/S2226585624000037>

**Government/Ecosystem****CPGRAMS**  
Government  
of India

- ❖ Centralized public grievance reporting, tracking, feed backend appeal.
- [pgportal.gov.in](http://pgportal.gov.in)

**MyGov**  
Government  
of India

- ❖ Citizen engagement through tasks, discussions, polls, campaigns and submissions
- [mygov.in](http://mygov.in)

**Atal  
Innovation  
Mission  
NITI-Aayog**

- ❖ Citizen engagement through tasks, discussions, polls, campaigns and submissions
- [aim.gov.in](http://aim.gov.in)

**Manthan —  
Principal  
Scientific  
Adviser**

- ❖ Multi-stakeholder collaboration across government, academia & industry.
- [psa.gov.in/manthan](http://psa.gov.in/manthan)